

ME 642: Computational modelling of granular flows

Course Code: ME 642

Course Name: Computational modelling of granular flows

L-T-P-C: 3-0-2-4

Prerequisites: Continuum mechanics / fluid mechanics / solid mechanics / soil mechanics

Intended for: MTech / PhD

Mutual Exclusion: NA

Approval: 60th BoA

Preamble

Granular materials are ubiquitous in nature and industry, spanning applications from landslides and avalanches to mineral processing and pharmaceutical manufacturing. Their complex flow behavior—exhibiting solid-like, fluid-like, and gas-like regimes—poses significant modeling challenges. This course, Computational Modelling of Granular Flows, offers a comprehensive foundation in theoretical, discrete, and continuum modeling approaches. Topics include particle-scale simulations using the Discrete Element Method (DEM), continuum formulations with depth-averaged models, and the Kinetic Theory of Granular Flows (KTGF). Coupled CFD–DEM techniques and emerging machine learning methods will also be covered. Students will gain hands-on experience with open-source software like LIGGGHTS, OpenFOAM, and CFDEM, enabling them to model and analyze granular flows in diverse engineering systems.

Course Modules with quantitative lecture hours

- **Introduction to granular flows and theoretical foundations:** Granular materials in nature and industry; particle properties: shape, size distribution, cohesion; flow regimes: quasi-static, dense, and dilute. Experimental characterization of granular flows; stress, strain, and constitutive models; flow instabilities, segregation; governing equations for particulate flows. (6 lectures)
- **Discrete element method:** DEM theory and contact models; normal and tangential forces, friction; time integration schemes, stability considerations; particle-wall interaction and boundary conditions; interaction with STL files; calibration and validation, hands-on with open-source LIGGGHTS code; applications: mixing and segregation, hoppers, chutes, vibrating screens. Principles of CFD-DEM coupling; inter-phase forces, multiphase flow applications, case studies in mineral processing and energy systems; use of open-source CFDEM code. (17 lectures)
- **Continuum modelling: depth-averaged model and rheologies:** Introduction to continuum modeling of granular flows; derivation of depth-averaged / shallow water mass and momentum equations; basal friction laws, source terms, Savage-Hutter model; numerical solution strategies for hyperbolic PDEs; analytical and numerical solutions and analyses; applications to landslides, avalanches and debris flows; demonstration of open-source codes and integration with digital elevation models; application of complex rheologies to dry and wet granular flows – including fluid-like (Newtonian, non-Newtonian), plastic (Mohr-Coulomb, Drucker-Prager) and intermediate (such as μ -I rheology). (9 lectures)
- **Kinetic theory of granular flows (KTGF):** Fundamentals of KTGF and application to dense granular flows; collisional dissipation (inelastic interactions); granular temperature equation; granular pressure; derivation of constitutive relations and types of granular viscosities; Eulerian two-fluid models using KTGF; applications in fluidized beds; hands-on with software (ANSYS Fluent, OpenFOAM) (6 lectures)

- **Advanced topics:** Handling non-spherical particles; cohesion; wet granular flows; machine learning applications in granular flows. (4 lectures)

Laboratory Modules

- **DEM:** Discrete element modelling in LIGGGHTS open-source code. Solve problems on mixing and segregation, hoppers, chutes, vibrating screens, and granular column collapse. (8 hours)
- **Depth-averaged modelling:** Simulation of geotechnical flows (landslides and snow avalanches) using depth-averaged continuum method in open-source and commercial codes. (6 hours)
- **KTGF:** Multiphase Eulerian modelling of pneumatic conveying in ANSYS Fluent. (4 hours)
- **CFD-DEM:** Modelling and simulation of gas-solid fluidisation and mixing in a vertical fluidised bed using open-source and commercial codes. (6 hours)
- **Machine learning:** Supervised ML using Scikit-learn, dataset preparation (using previous lab results), regression and classification. (4 hours)

Text Books

1. Rao, K. Kesava, Prabhu R. Nott, and S. Sundaresan. *An introduction to granular flow*. Vol. 490. Cambridge: Cambridge university press, 2008.

References

1. O'Sullivan, Catherine. *Particulate discrete element modelling: a geomechanics perspective*. CRC Press, 2011.
2. Thornton, Colin. *Granular Dynamics, Contact Mechanics and Particle System Simulations: A DEM study*. Springer, 2016.
3. Pudasaini, Shiva P., and Kolumban Hutter. *Avalanche dynamics: dynamics of rapid flows of dense granular avalanches*. Springer Science & Business Media, 2007.
4. Gidaspow, Dimitri. *Multiphase flow and fluidization: continuum and kinetic theory descriptions*. Academic press, 1994.
5. Norouzi, Hamid Reza, et al. *Coupled CFD-DEM modeling: formulation, implementation and application to multiphase flows*. John Wiley & Sons, 2016.